Algorithms, Machine Learning, and Collusion*

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Abstract

This paper discusses the question whether self-learning price-setting algorithms are able to coordinate their pricing behaviour to achieve a collusive outcome that maximizes the joint profits of the firms using these algorithms. While the legal literature generally assumes that algorithmic collusion is indeed possible and in fact very easy, the computer science literature on cooperation between algorithms as well as the economics literature on collusion in experimental oligopolies indicate that a coordinated and in particular tacitly collusive behaviour is in general rather difficult to achieve. Many studies have shown that some form of communication is of vital importance for collusion if there are more than two firms in a market. Communication between algorithms is also a topic in artificial intelligence research and some recent contributions indicate that algorithms may learn to communicate, albeit in a rather limited way. This leads to the conclusion that algorithmic collusion is currently much more difficult to achieve than often assumed in the legal literature and is therefore currently not a particularly important competitive concern. In addition, there are also several legal problems associated with algorithmic collusion, for example, questions of liability, of auditing and monitoring algorithms as well as enforcement. The limited resources of competition authorities should rather be devoted to more pressing problems as, for example, the abuse of dominant positions by large online-platforms.

JEL–Classifications: C45, D43, K21, L41

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1 Introduction

The increasing importance of Big Data in the digital economy and the use of price-setting algorithms in many industries have raised concerns that these new technologies may cause some novel competition problems. One of the issues that is discussed intensely in the competition law and policy community is whether smart, self-learning price-setting algorithms might facilitate or even cause collusive behaviour in oligopolistic markets. It is suspected that self-learning algorithms, by observing the pricing behaviour of algorithms employed by their competitors, could learn within a short time to coordinate their prices and behave like a cartel to maximize the joint profit of the firms employing such algorithms. It is suspected that this type of

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"algorithmic collusion" may occur not only in markets with a few participants but also in markets with a larger number of firms, so-called wide oligopolies. This new type of collusive behaviour gives rise to some new problems with respect to competition law, in particular in the context of liability issues, with respect to the definition of an agreement, the monitoring of algorithms and the enforcement of competition law in case of algorithmic collusion.

These concerns have been raised mainly by legal scholars who have pointed out that smart, self-learning algorithms might be able to quickly learn to play a coordinated equilibrium in price setting oligopolies. Since 2015, several papers on these suspected negative impacts of algorithms on competition have been published as, for example, the contributions by Ezrachi/Stucke (2015, 2016a, 2016b, 2017), Mehra (2015), Oxera (2017), Pasquale (2016), Woodcock (2017) among others. This discussion was also featured in the popular and economic press as well as in antitrust journals. Reports on the dangers of algorithmic collusion have been published in the Economist, the New Yorker, in the Harvard Business Review, the Financial Times and have caused substantial publicity of this topic.[1] The Antitrust Chronicle has devoted a whole issue on the concerns related to the effects of algorithms on competition.[2] Many competition authorities, including the Directorate General of Competition of the European Commission, the German and the French competition authorities seem to take this issue very seriously and also the German Monopolies Commission has discussed this topic in its recent report. Algorithmic collusion was also discussed at several competition policy conferences, e.g. at the International Cartel Conference in Berlin, an OECD Roundtable (OECD (2017)), but also at academic conferences as the Cresse Conference on industrial organization or the Ascola conference 2018. There is, however, a debate whether the concern that self-learning, price setting algorithms are indeed able to behave in a coordinated way and charge inflated prices like a monopolist is justified or whether this discussion belongs to the realm of "legal sci-fi" (Schrepel (2017)).

There are at least two aspects with respect to the discussion of algorithmic collusion that are not completely satisfactory. First, in the contributions by legal scholars on the competitive effects of price setting algorithms, they are treated as a mysterious black box – it is not explained how these algorithms work, what machine learning is, which types of machine learning are employed in price setting algorithms, how algorithms learn and what they are able to learn. In general, these contributions just assume that it is quite easy for autonomous price setting algorithms to behave in a coordinated way. The question, under which conditions in a multi-agent framework coordination can be achieved, is not discussed. Secondly, these contributions do not refer to the theoretical and experimental economic literature on tacit collusion. Here, it was demonstrated that it is generally not easy to achieve a tacitly collusive, profit maximizing outcome. In particular, it has turned out that communication between agents is of vital importance for a collusive outcome to be reached if there are more than two firms in a market.

In this paper the concerns expressed in the legal literature about algorithmic collusion are examined in the light of several contributions from the computer science literature on machine learning and algorithmic coordination in multi-agent settings. This literature indicates that coordination is in general not easy to achieve and depends strongly on the specific setting that is used and also tends to become more difficult when the complexity of the algorithms

[1] Price-Bots Can Collude Against Consumers, in: *Economist*, May 6, 2017; When Bots Collude, in: *The New Yorker*, April 25, 2015; How Pricing Bots Could Form Cartels and Make Things More Expensive, in: *Harvard Business Review*, October 27, 2016; Policing the Digital Cartels, in: *Financial Times*, January 8, 2017.

[2] The Algorithms Have Landed! in: *Antitrust Chronicle*, May, Volume 2, Spring 2017.

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increases. As the algorithms employed in practice are increasingly sophisticated due to the rapid progress in machine learning, this tends to reduce the probability of collusion. This result corresponds to the findings of the experimental economics literature which has shown that collusive behaviour in case of more than two firms in the market generally occurs only if they are able to communicate. Therefore, the question is discussed whether algorithms can learn to communicate as this seems to be a vital requirement for collusion. Some recent papers show that algorithms may indeed be able to learn to communicate, albeit in a very limited and restricted way. Thus, on the one hand for algorithms with a higher degree of complexity, coordination becomes more difficult, but on the other hand, more sophisticated algorithms may be able to learn to communicate with each other, making coordination more likely. Given the current state of research in artificial intelligence and machine learning, the concerns with respect to algorithmic collusion do not seem to be justified at the moment. Considering the rapid progress in AI-research, however, it cannot be excluded that in the future, algorithms may learn to communicate and to behave in a collusive way. Therefore, the question is considered whether any changes in competition law due to the issue of algorithmic collusion is required and if so, which changes would be advisable.

The paper is organized as follows: In section two the arguments concerning the danger of algorithmic collusion as given in the legal literature are presented. In section three, a brief introduction to the different types of machine learning, i.e. supervised, unsupervised and reinforcement learning, is provided. In section four some results of the game theoretic as well as the AI-literature about algorithmic coordination are discussed. Section five contains a brief summary of the results of the experimental economics literature on tacit collusion. It turns out that communication between agents is of central importance for a collusive equilibrium to emerge if there are more than two firms. Section six briefly surveys the literature on algorithms learning to communicate and it is shown that algorithms might indeed learn to communicate with each other, albeit in a rather limited way. Some suggestions to change the current competition law to deal with the problem of algorithmic collusion are discussed in section seven. Section eight concludes.

2 Algorithms as collusive device

In the context of firms using price setting algorithms, two major competitive problems are discussed in the literature. First, by employing algorithms in combination with Big Data on consumer preferences, buying habits, search histories etc., firms may be able to charge personalized prices that allow the firms to exploit the consumers' willingness to pay and thereby to increase their profits. The second competition concern is related to the impact of algorithms on collusive or coordinated behaviour in markets with imperfect competition. In what follows, the focus is on the second concern, the issue of algorithmic collusion.

2.1 Algorithms and structural market conditions

With respect to algorithms and collusion, two different dimensions have to be considered how algorithms may affect collusive behaviour. First, algorithms may change the structural market conditions in such a way that collusion becomes easier. For example, if firms employ

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algorithms that collect and analyze prices charged by competitors, market transparency is increased, deviations from a collusive price can be detected faster and more easilyu. Also, punishment mechanisms may become more effective as prices can be changed very quickly and such a repricing allows for a swift retaliation if a deviation from a coordinated equilibrium occurs.

It is however, doubtful whether algorithms create or increase barriers to entry. Many firms that employ pricing algorithms are online-platforms characterized by network effects and economies of scale. This tends to increase the volume and the velocity of the data collected and could thus lead to improved pricing algorithms, making it more difficult for newcomers to enter such a market, i.e. erecting a "data barrier to entry". On the other hand, there are many independent firms that offer algorithmic pricing as a service, e.g. Feedvisor, PricingPro or Intelligence Node, which makes market entry easier by converting the fixed cost of installing data analytics capacities into variable cost.

In markets that are characterized by competition through innovation, improved algorithms yield a competitive advantage, for example by providing consumers with personalized offers and products which tends to reduce the scope for collusion. Also, efficient algorithms could lead to cost advantages, for example in the context of logistics. There seems also to be a tension between collusion on the one hand and personalized pricing on the other because in markets where personalized pricing is possible, most of the consumer surplus can be extracted which decreases the incentives to collude. Thus, it may be conjectured that the incentives to collude are particularly high when personalized pricing and personalized offers are not possible as for example in cases of standardized products.

Also, there are some efficiency advantages related to the use of algorithms. For example, changing prices becomes easier as no human action is required and thus prices can be more easily adjusted to changing demand conditions, thereby increasing market efficiency. As market transparency is increased, better information on actual prices is is available for consumers as well. This enables price comparison sites to provide consumers with almost real time prices, information about sales offers etc.[3] Also, algorithms could be employed by competition authorities to monitor markets and to detect unusual pricing behaviour, anomalies and can scan for competitive problems, e.g. cartels or collusive behaviour.

Whether a widespread use of algorithms changes structural market conditions in favour of collusion or whether they increase the competitive pressure is thus not obvious and will in general depend on the case at hand. In some markets, conditions might be changed by algorithms so that collusion becomes less likely – in other markets, the opposite might hold true.

2.2 Algorithms as a facilitating device

Algorithms may not only change the structural market conditions but may be employed directly to facilitate collusion, i.e. algorithms may be considered as a so-called facilitating device. For example, algorithms can be used by firms as an additional instrument to coordinate a cartel. In this context, the Topkins and the Trod cases could be mentioned.[4] Here, online firms

[3] see e.g. Gal/Elkin-Koren (2016).

[4] https://www.justice.gov/atr/case/us-v-david-topkins; https://www.justice.gov/atr/case/us-v-daniel-william-aston-and-trod-limited.

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selling posters on Amazon Marketplace agreed on using the same algorithm to fix the prices of their products. Another case where price-setting algorithms have been used as a coordinating device was the Eturas Case where a booking system was employed as a tool for coordinating the actions of the firms.[5]

With respect to the direct impact of algorithms on competition, Ezrachi/Stucke (2015, 2016) have suggested several scenarios to categorize the different functions an algorithm may have with respect to the coordination of firms' activities. The first scenario is the *messenger scenario*. Here, firms use algorithms to implement a cartel as for example in the above mentioned Topkins and Trod cases. They agreed to employ algorithms that ensure that the cartel price is charged also if market conditions change. This however, does not create any competitive concerns of a new quality as in these cases an explicit agreement between the firms is necessary. Algorithms are only a new technology to organize a cartel.[6] Cartelists have and always will use the available technology to form a cartel – be it telephone, email, computer programmes etc. This is nothing new or special to algorithms. It is often claimed when a new technology is introduced that changes in competition law are necessary – in general, however, this is not the case. The technology used to implement a cartel may be different but this does not necessarily imply that competition law has to be changed. Mostly, the existent competition law is fully sufficient to deal with this type of competition problem.

Slightly different from the messenger scenario is the *hub-spoke scenario*. Here, several firms use the same algorithm for example by outsourcing pricing decisions to the same third party that offers algorithmic pricing as a service as e.g. Boomerang or Pricing Pro. This third party would in fact act as the hub in a hub-spoke cartel. It is suspected that a coordinated equilibrium could arise if the price setting by a central algorithm leads to prices that maximize joint profits. It is, however, unclear whether the "hub" employs differentiated specialized algorithms for different customers or what would happen if firms in the same relevant market connect to different price-optimizing firms, e.g. Feedvisor, PricingPro and Boomerang. As these firms are competing against each other, and each one has an incentive to provide the superior solution for its customers. This implies that the algorithms they provide will in general differ, observe different things and thus learn differently. Also, many algorithms contain stochastic elements to explore the environment. It is thus doubtful whether these algorithms will converge to a stable set of (profit maximizing) prices. If firms differ with respect to their technology and thus their cost function, a solution that maximizes the joint profit of all "spoke" firms might not be achievable. Also, for a "hub-spoke" scenario to emerge, some communication or agreement is necessary for the firms to coordinate on which hub to use. This type of agreement would amount to a common case of a hub-spoke cartel and would therefore not raise any new competition law problems. To quote FTC Commissioner Ohlhausen: If the word "algorithm" can be replaced by the phrase "a guy named Bob", then algorithms can be dealt with in the same way as in a traditional hub-spoke cartel.[7]

The third scenario, called the *predictable agent* is characterized by each firm developing its algorithm unilaterally to deliver predictable outcomes and react in a given way to changing market

[5] C-74/14 - Eturas

[6] It has to be taken into account, however, that algorithms may open new ways to facilitate collusion, for example by secretly exchanging information between firms which may be difficult to detect by competition authorities. See Gal (2018).

[7] Ohlhausen (2017).

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conditions, with awareness of likely developments of other algorithms used by its competitors.[8] As algorithms monitor the price setting by other algorithms and follow price increases, an interdependence between algorithms is created. Algorithms can react very quickly, i.e., if one firm lowers the price, other firms lower their prices immediately. If this is the case, competitive pressure is reduced and collusive behaviour becomes more likely. This working of algorithms is similar to a price guarantee that also decreases competitive pressure as they reduce the incentives to increase demand, market shares and profits by undercutting the price of a competitor. Since algorithms can react very quickly, there is hardly any time during which the price of a competitor can be undercut to increase market share and gain additional revenue and profit. Also, similar to the case of price guarantees, all firms have an incentive to use price setting algorithms. It is suspected that an industry-wide independent use of algorithms by firms could lead to a collusive outcome. This however, does not cause any novel competitive problems that would not be present as for example in the case of a widespread use of price guarantees. It is, however, unclear, what happens if firms use different algorithms that react slightly differently to market conditions. An example of what may happen if different algorithms are employed is the case of the genetics textbook "The Making of a Fly" that was offered by two sellers on Amazon Marketplace. One seller used the algorithm: "Charge a price that is 1.27059% times the average price of the competitors", while the other employed the rule: "Set a price that is 0.9983 times the lowest price of any competitor." The interaction of these simple "algorithms" led to a price of the book of more than than 23 Mio.$ – of course, no copies of this book were sold at this price.[9] But even if firms would use identical algorithms, each one would probably observe different things and also learn differently as algorithms often often contain stochastic elements. This raises serious doubts whether even in the predictable agent scenario prices will result that are above the competitive level or even maximize joint profits.

2.3 Collusion between algorithms

The most interesting scenario described by Ezrachi/Stucke (2015, 2017) is the *autonomous machine scenario*. It differs with respect to the former three scenarios as here the algorithms are not mere instruments used by humans to facilitate a cartel, but are autonomous agents created by the companies to achieve a given target, such as profit maximisation: "The machines, through self-learning and experiment, determine independently the means to optimise profit."[10] The authors claim that, also due to the Internet of Things, firms would be perfectly informed not only about their own logistics, production, customers, sales, deliveries and so on, but would also know everything about their competitors, the market, the consumers etc. Stucke/Ezrachi (2016) claim that these algorithms would possess a "God view". Using all this information, self-learning, smart algorithms would easily arrive at a collusive outcome by avoiding the "death spiral" of the "Tit-for-Tat"-strategy.

The authors claim that the monitoring capabilities of the algorithms are so powerful that tacit collusion would come about very quickly: "Two Artificial Neural Networks and one Nash equilibrium meet in an online (pub) hub. After a few milliseconds, a unique silent friendship is

[8] Erzrachi/Stucke (2015).

[9] Supposedly, there were also cases were algorithms drove down the price to or even below marginal cost, for example if both firms would have employed the second pricing rule. These cases are, however, not discussed in the press.

[10] Erzrachi/Stucke (2015), 9.

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formed . . . " (Ezrachi/Stucke (2017)). In addition, this would lead to cartel-like outcomes also in markets with a large number of firms: "With the industry-wide use of computer algorithms, we may witness conscious parallelism in markets with many more players, where collusion previously would have been unstable."[11] All in all, the legal literature assumes that achieving such a tacitly collusive, profit maximizing equilibrium is very easy or even unavoidable.

It is, however, unclear whether the autonomous agent scenario will indeed lead to this type of collusive outcome. It highly depends on what these algorithms are capable of, whether they can indeed coordinate their behaviour or even can learn to communicate with each other. To explore these questions in more detail, the next section gives a brief survey on the topic of machine learning and sketches the different types of machine learning documented in the AI-literature and applied in AI-research. Also, the relevant literature in computer science about algorithms learning to coordinate their behaviour in particular in social dilemma scenarios as the prisoner's dilemma is presented.

3 Algorithms and machine learning

First, it should be pointed out that an algorithm is but a fancy term for what was formerly known as software or a computer program. More formally, an algorithm ". . . is an unambiguous, precise, list of simple operations applied mechanically and systematically to a set of tokens or objects. (. . .) The initial state of the tokens is the input; the final state is the output."[12] Usual properties of an algorithm are finiteness, i.e. it must always terminate after a finite number of steps, and definiteness, that is, each step must be precisely defined and the actions to be carried out must be rigorously and unambiguously specified for each case.

The algorithms that are employed by firms to set prices vary significantly with respect to the degree of complexity. Some are just very simple heuristic rules of thump as for example the simple finite automata considered in the early game theoretic contributions to the analysis of boundedly rational behaviour. At the other end of the spectrum there are much more complex pricing algorithms based on deep neural networks that are also used for example in self-driving cars, natural language processing, image recognition or automatic translation etc. These algorithms can modify themselves over time, depending on the input they receive and this modification can be considered as learning. The different types of machine learning that are considered in the AI literature will be discussed briefly in the next section.

3.1 Machine learning

Machine learning is a field of computer science, in particular of artificial intelligence (AI) that can be described as general purpose approaches to learn functional relationships from data without the need to define these relationships a priori, i.e. without being explicitly programmed. A commonly used definition of machine learning has been suggested by Mitchell (1997): "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."[13] In general, three different types of machine learning are con-

[11] Ezrachi/Stucke (2016b), 77.
[12] Wilson/Keil (1999), 11.
[13] Mitchell (1997), 2.

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sidered in the AI-literature: supervised (and semi-supervised) learning, unsupervised learning and reinforcement learning. In what follows, these different types of machine learning are briefly discussed.

3.1.1 Supervised learning

In *supervised learning*, an algorithm is presented with example data and associated target values in order to later predict the correct target values after the training when confronted with new examples. When these predictions turn out to be incorrect, the algorithm is modified. This training process continues until the model achieves a desired level of accuracy on the training data. Stated otherwise, from a set of examples a general rule is derived.

There are basically two types of tasks that an algorithm is trained for in supervised learning: regression analysis and classification. In a regression, the output is a real number (or a vector), e.g. the prediction of the price of a house. The algorithm has been trained by providing a set of data on houses, for example features like size, floors, bedrooms, location etc. and the respective prices to accurately predict the price of a house with certain features.

In a classification task, the inputs are divided into two or several classes, e.g. a tumour being malignant or benign or an email being spam or no-spam. The output is a class label, in the simplest case it is either 0 (spam) and 1 (no-spam). Here, the training set consists e.g. of a set of words associated with spam emails (offer, deal, bargain etc.) and the algorithm is trained to learn to classify emails into spam and no-spam. Classification is also used for pattern recognition, for example the recognition of handwritten numbers or clustering customers into different groups.

3.1.2 Unsupervised learning

A different form of machine learning is *unsupervised learning*. Here, a model is prepared by deducing structures or patterns present in the input data to extract general rules without any associated response. An algorithm restructures data for example by organizing them by similarity. Here, a new pattern emerges with clusters formed by similar examples while examples in distinct clusters are dissimilar. Since the examples provided to the algorithm are unlabelled, there is no evaluation of the accuracy of the structure that is the output of the algorithm. Examples of unsupervised learning are organizing news into different categories, for example politics, finance, sport etc. or recommender systems were the recommendations a customer receives are based on an estimation of what group of consumers this customer resembles the most. This is then used to draw conclusions about the customer's preferences and to generate appropriate recommendations. As the detection of hidden structures and patterns yield additional insights into the meaning of the data, the outcome of unsupervised learning can be a useful input into supervised learning algorithms.

3.1.3 Reinforcement learning

Finally, there is the type of machine learning that is of particular interest in the context of price setting by firms which is known as *reinforcement learning*. Here, algorithms learn to take actions in an (unknown but fixed) environment so as to maximize some notion of cumulative

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reward. The output is an action or sequence of actions and the only supervisory signal is a (scalar) reward. In general, under reinforcement learning, an algorithm is not trained which action is the best in a given situation, but it receives an occasional reward if a successful action has been chosen. The objective is to select actions such that the expected sum of the discounted future rewards is maximized.[14]

Reinforcement learning differs from standard supervised learning in that neither correct input/output combinations, such as a house with certain characteristics and its price are presented, nor actions that have proved unsuccessful are explicitly corrected. Instead, the emphasis of reinforcement learning is on developing successful strategies that maximize long-term returns. When choosing an action, it is also important to find a balance between exploring the partly unknown environment and exploiting the available information. The algorithm learns by trial and error, because a behaviour that has proved unsuccessful is punished, but successful actions are rewarded. Unsuccessful strategies are therefore used less frequently, while successful strategies receive a positive feedback because they promise higher expected payouts. In order to achieve the best possible result, the algorithm must take into account the long-term consequences of its actions. This may imply that to maximize the long-term payoff, short-term sanctions have to be accepted.

The algorithm is characterized by a function V that evaluates a certain state s_t (state-value function). In order to improve its strategy π it approximates V to the ideal V^π. The simplest algorithm selects an action based on its strategy in each iteration, observes the reward and adjusts the evaluation function according to the following rule:

$$V(s_t) \leftarrow V(s_t) + \alpha \left(r_{t+1} + \delta V(s_{t+1}) - V(s_t)\right),$$

where α is the learning rate and δ denotes the discount factor. There are different types of reinforcement learning, for example, Monte-Carlo algorithms and temporal-difference (TD) learning. An agent with a TD learning algorithm does not make the adjustment when he receives the reward, but after each action based on an estimated expected reward. Two often employed variants of TD-learning are Q-Learning and SARSA-Learning.

In Q-Learning the agent evaluates the utility of an action instead of a state. The learned function $Q(s, a)$ is therefore called action-value function and ultimately gives the expected utility of taking a given action a in a given state s, and following an optimal policy thereafter. The algorithm executes an action a according to his strategy for the current state s and receives the resulting reward r. From the successor state s', the agent chooses the most promising action a' according to its current action-value-function as a future reward. It uses this to adjust the action-value function according to the following rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(r + \delta \max_{a'} Q(s', a') - Q(s, a)\right).$$

SARSA stands for State-Action-Reward-State-Action and is also an algorithm for learning an action-value function. In contrast to Q-Learning, however, the agent sticks to his strategy when calculating his follow-up action. An agent who implements SARSA carries out an action a according to his strategy for the current state s and receives a reward r. In the subsequent state s', he now selects an action a' according to his strategy and takes its valuation as a future

[14] For a comprehensive introduction to reinforcement learning see Sutton/Barto (2017).

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profit to adjust the action-value-function according to the following rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left(r + \delta Q(s',a') - Q(s,a)\right).$$

Besides the three types of machine learning, supervised, unsupervised, and reinforcement learning, there is also the concept of *deep learning*.

3.1.4 Deep learning

In contrast to the different variants of machine learning, deep learning does not refer to the type of learning but the type of computer program that is employed, so called artificial neural networks (ANN). Neural networks are computer programs that try to mimic the function of the brain by learning from experience and understanding the world in terms of a hierarchy of concepts, with each concept defined through its relation to simpler concepts. Complicated concepts are learned by a computer by building them out of simpler ones. (Goodfellow et al. (2016), 1).

An ANN consists of several layers of connected artificial neurons, an input layer, an output layer and one or more hidden layers. The neurons of the input layer receive input data and change their inner state, i. e. they are "activated" when a predefined threshold value is exceeded by the input data and then send an output to the neurons of the first hidden layer, which then modify this data according to a learning rule. This depends on the input, the activation, the connections between the neurons as well as the "weight" of each connection and forward it to the next hidden layer, etc. The input data is thus gradually processed and transferred to the other hidden layers, where it is modified by the neurons of the next layer. A simple ANN can be illustrated as follows:

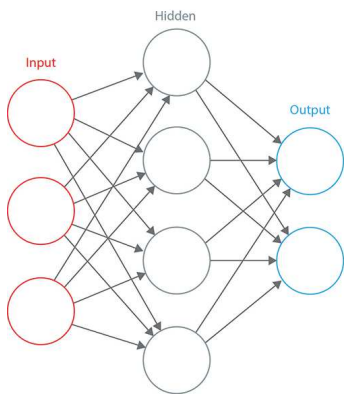

Figure 1: A simple neural network with one hidden layer

Depending on the received data and the activation, the neural network "learns" by modifying the connections between the neurons and the threshold values. This learning can be supervised, unsupervised or reinforcement learning. The result of the computation is finally provided by the neurons of the output layer.

While neural networks are a quite old concept, the first examples consisted only of a few neurons and a few layers because of limited computational and storage capacities. Only recently ANNs with many layers could be implemented due to the availability of high performance computing with graphic processing units (GPU) that make parallel computing faster and cheaper,

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almost unlimited storage capacities and huge volumes of available data. These deep neural networks (DNNs) today typically have a few thousand to a few million units and millions of connections. For example, in the Google Brain project more than 16000 computers are connected with more than 1 billion connections. Learning by DNNs is called "deep learning" which has important applications for example speech recognition, natural language processing (automatic translation), direct marketing, recommender systems. DNNs are able to learn complicated non-linear relationships, e.g. to play Go (AlphaGoZero) or Poker (Libratus) by playing against themselves a large number of games.[15].

The next section gives a brief survey of the literature on the interaction and learning of algorithms in strategic, multi-agent environments which are of particular importance in the context of oligopolistic coordination where several firms are present and interact with each other in a market. In these scenarios, the main problem consists in the non-stationarity of the environment that is generated by the presence of other agents that also learn and adapt to the changing environment. This amounts to the problem of learning a moving target. In what follows, the focus is on social dilemma situations as for example the prisoner's dilemma, because these strategic scenarios are closely related to the problem of coordination in an oligopolistic markets.

4 Algorithmic coordination in the literature

There are different strands in the economic literature that have investigated the problem of coordination in oligopolistic markets or in similar strategic situations, i.e. in prisoner's dilemma games. The theoretical approaches often analyze collusion in oligopolies in the framework of repeated games since interaction between the players on markets usually occurs repeatedly over time. Here, however, the problem arises that in complex games with many players and complex strategy spaces, equilibria cannot be efficiently computed and convergence to an equilibrium can take a very long time.[16] In addition, even in the simplest two-player, two-strategy repeated prisoner's dilemma, the folk theorems of game theory show that there are uncountably infinitely many Nash-equilibria (an "embarrassment of riches" (Tirole (1989)), that span the whole spectrum of possible types of behaviour between full coordination at one extreme and no coordination at the other. In addition, it is not clear how players chose an equilibrium. Game theory makes statements about the existence and the properties of Nash-equilibria, but not on how or whether these equilibria are achieved by actual players.

Therefore, it has been tried to solve the problem of equilibrium selection in games by relaxing the assumption of the players' perfect rationality, as it has been suspected that there are so many possible equilibria mainly because the players are "too rational". Thus, boundedly rational behaviour was considered and complemented by the possibility of learning. There are several contributions that use finite automata, which can be considered as very simple types of algorithms, to model bounded rationality. The third strand of literature comprises the contributions of experimental economics that investigate the behaviour of human subjects playing an oligopoly game in a computer lab. The next section considers the contributions

[15]For the development and the performace of AlphaGoZero see Silver et al. (2017), for the algorithm Libratus see Hsu (2017).

[16]Computing equilibria in complex games can be NP-hard, i.e. such games cannot be solved in polynomial time. See e.g. Daskalakis et al. (2009).

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from economic theory that analyze the outcomes if finite automata play the repeated prisoner's dilemma. Also, some related models of learning in games are presented.

4.1 Algorithms, learning and repeated games

The first models of algorithms playing a repeated prisoner's dilemma date back to the mid-1980s. These models have been developed in the context of analyzing boundedly rational behaviour where players are restricted to carry out their strategies by very simple algorithms known as finite automata.[17] A finite automaton consists of a set of states, an input and an output function as well as a transition function. The automaton receives an input, possibly changes its internal state according to the transition function and sends an output. It was demonstrated that in this framework the cooperative outcome is in general not an equilibrium. This is mainly due to the assumption that each internal state of an automaton is costly. Therefore, only automata without redundant states are used in equilibrium. Tit-for-Tat (and also Grim-Trigger) strategies do not form an equilibrium as the punishment states are not used in equilibrium. A somewhat different approach to model bounded rationality has been followed by Cho (1994) who considers a simple neural network and shows that the perfect folk theorem can be recovered with neural networks hat contain only a small number of neurons, i.e. also in this case all possible outcomes can be an equilibrium in the repeated game.

The research on bounded rationality in economics has pointed out that even players that are not fully rational are nevertheless able to learn form past experience. Therefore, this aspect has subsequently been recognized in the literature on finite automata. The simple algorithms that had been considered in the early contributions were modified to include some forms of learning which became an important research topic in the context of boundedly rational behaviour. The research on learning in games has led to a number of important contributions about different types of learning in games.[18] For example ficticious play, replicator dynamics from evolutionary game theory, adaptive learning, imitation etc. have been analyzed. Also in the context of finite automata, different concepts of learning have been explored. For example Miller (1996) uses a genetic algorithm to analyze whether cooperation could be the outcome in a repeated prisoner's dilemma In his model, a genetic algorithm randomly recombines binary coded strategies where also some random modifications (mutations) occur. Successful strategies, i.e. strategies that yield a high payoff, have a higher probability of being passed on to the next generation of strategies. This process attempts to model the evolutionary process of recombination, mutation and selection. It turns out that this type of replicator dynamics leads to a collusive outcome in a two-player prisoner's dilemma played by finite automata. The degree of cooperation that can be achieved however, depends on the degree of "noise", i.e. the misreporting of the other player's moves. Miller demonstrates that if the moves of the opponent cannot be perfectly observed, only very limited degrees of cooperation will arise.

Cooperation in a repeated prisoner's dilemma may also result if the agents are characterized by satisficing learning as has been shown by Stimpson et al. (2001). This type of learning assumes that in each period an agent compares his payoff with an aspiration level – if the payoff is at

[17] See e.g. Rubinstein (1986) and Abreu/Rubinstein (1988), Banks/Sundaram (1990). For a survey on this literature see Marks (1992).

[18] For a detailed and extensive discussion of the different aspects of learning in games see Fudenberg/Levine (1999).

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least as high as the aspiration level, the action played in the last period is repeated in the next period. Otherwise, the aspiration level is updated and the new aspiration level is calculated as a weighted average of the old level and the actual payoff received. Simulations have shown that in most cases, this type of learning converges to the one-shot Nash equilibrium, but agents occasionally learn to cooperate.

4.2 Reinforcement learning in games

Since recent approaches of machine learning by autonomous agents generally apply reinforcement learning which is also the type of learning that is considered in the literature on algorithmic collusion, the focus of this section is on applications of reinforcement learning in the game theory and computer science literature.

In the game theoretic literature, the machine learning approach of reinforcement learning seems to have first been employed by Roth/Erev (1995) to explain experimental data by showing that reinforcement learning tracks the observed choices. Also, reinforcement learning has been used in a paper by Börgers/Sarin (1997) where this type of learning is compared with the replicator dynamics used in evolutionary game theory. Here, a population of players is considered where each player uses one of finitely many (pure) strategies. Players are repeatedly pairwise randomly matched and play against each other. The proportion of players characterized by a strategy that yields a payoff that is larger than the average payoff in the population is increased in the next period. Learning thus relates to a whole population of players and not to individual players. Börgers/Sarin find that under certain conditions, reinforcement learning and the replicator dynamics of evolutionary game theory both converge to the same equilibrium. These papers, however, do not study the question whether reinforcement learning may lead to cooperation in repeated prisoners dilemma or in oligopoly games.

In an early paper that looks at different learning types in an oligopoly experiment (Huck et al. (1997)), reinforcement learning is not considered as particularly helpful: "In particular, information about the market and information about the individual play of others turned out to be very important treatment effects [...]. Neither effect can be explained by reinforcement learning."[19] In two papers by Erev/Roth (1999, 2001), this type of learning is applied to repeated prisoner's dilemma games and they show that simple reinforcement learning will in general not lead to a cooperative outcome. It has been demonstrated later in a more general framework (Beggs (2005)) that the type of reinforcement learning employed by Erev/Roth (1999, 2001) and Börgers/Sarin (1997) generally converges to the one-shot Nash equilibrium. Another model that employs a simple reinforcement learning algorithm to a finite automata framework is Hanaki et al. (2005). They consider all 26 strategies that can be implemented by a two-state automaton and consider a two-phase learning process. In the first phase, the players explore the environment, i.e. the different strategies and limit their attention in phase two to those strategies that have been successful in phase one. Simulation results show that in the framework of strategies with limited complexity, cooperation is the predicted outcome in the repeated prisoner's dilemma. The early game theoretic literature does not seem to be particularly supportive with respect to the emergence of cooperation under reinforcement learning.

[19] Huck et al. (1997), 22.

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4.2.1 Reinforcement learning and collusion in computer science

There is a large literature in computer science on reinforcement learning and cooperation in repeated prisoner's dilemma games that seems not to be very well known among competition economists. Because there are so many studies, only a few representative contributions can be considered.[20] An early contribution from computer science that applies reinforcement learning in a multi-agent setting is Tan (1993) who shows that communication between agents about the actual as well as past actions and rewards or about learned policies, improves the outcome of each agent. This result already hints at a connection between learning, communication and cooperation in games because in this model, communication between agents makes cooperation easier.

There are several papers that consider different variants of Q-learning, a special type of reinforcement learning, in prisoner's dilemma situations (Banerjee/Sen (2007), De Cote et al. (2008), Kaymak/Waltman (2006), Moriyama (2007, 2008)) The results of these studies show that cooperative behaviour is, under certain assumptions about the memory of the agents, the learning rate and the specific values of the payoffs or the information players have about other players' payoffs etc., a possible outcome of a reinforcement learning process but is by no means the only result.[21] As it turns out, the specific features of the assumed learning algorithm are of central importance with respect to which type of equilibrium the algorithms converge. For example, if a variable learning rate is assumed that is characterized by a low learning rate if the agent receives a high payoff (Win) and a high learning rate in case of a low payoff ("Win-or-Learn-Fast" (WoLF)), the learning process usually does not converge to a cooperative outcome. However, if a variable learning rate is employed that uses the principle of a high learning rate if the payoff changes and a low learning rate otherwise ("Change-or-Learn-Fast" (Colf)), Q-learning converges to a cooperative outcome. Convergence to the one-shot Nash-equilibrium, however, is still a common outcome of many Q-learning models in a multi-agent setting.[22]

In the last couple of years, significant progress has been made with respect to self-learning algorithms due to the increasing computational power and the storage capacities on the one hand and the rapid progress in artificial intelligence, machine learning, and deep learning on the other. Two recent papers that analyze the problem of algorithmic coordination in social dilemma situations, i.e. strategic situations characterized by a tension between collective and individual rationality as for example the prisoner's dilemma, the stag hunt game or the game of chicken, using deep reinforcement learning are Leibo et al. (2017) and Crandall et al. (2017).

In Leibo et al. (2017) different social dilemma situations in complex multi-agent environments are considered that are characterized by a payoff structure similar to the payoffs in a prisoner's dilemma. The agents are modelled as deep neural networks and learn to play two different sequential social dilemma situations, a gathering and a hunting game. In the gathering game, the players have to collect resources and each player has the option to 'tag', i.e. temporarily remove another player. It turns out that players learn to cooperate, i.e. don't tag other players,

[20] For a survey of the early contributions to reinforcement learning and its relation to game theory, see Shoham et al. (2007) and the comments by Fudenberg/Levine (2007).

[21] Usually, a so-called self-play is considered, i.e. all agents are characterized by the same algorithm.

[22] For a survey of the outcomes of different algorithms with respect to cooperation, see Crandall/Goodrich (2011). It should be mentioned also that the objective of many of the computer science models of reinforcement learning is not the explanation under which conditions a cooperative outcome will result, but rather the problem how a machine learning algorithm should be designed to ensure a cooperative outcome.

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if resources are plentiful while, if resources are scarce, players behave non-cooperatively by trying to temporarily remove other players. In the hunting game, two players (the "wolves") have to hunt (touch) a third player. If either wolf touches the prey, the other wolves receive a reward that depends on all wolves in the neighbourhood. If, however, two wolves capture the prey together, they receive a higher payoff because they may be better able to defend the prey against scavengers. It turns out that learning to cooperate in the hunting game is more difficult than in the gathering game as in the latter the players need only be concerned with the resources and may not depend on the other player's actions. In the hunting game, cooperation needs more coordination as actions have to be conditioned on the presence of a partner.

The paper by Crandall et al. (2017) deals with the construction of algorithms that are capable of forming cooperative relationships with other algorithms but also with humans. These algorithms should exhibit a superior performance in a wide variety of scenarios, should be able to establish relationships with other algorithms or humans without prior knowledge of their behaviour, should be able to deter exploitative behaviour and determine how to elicit cooperation from a (potentially distrustful) partner who might be disinclined to cooperate. In addition, the algorithm should exhibit a high learning speed. Crandall et al. consider 25 different algorithms, including classic strategies like tit-for-tat, win-stay-lose-shift, evolutionarily stochastic strategies, machine-learning algorithms (including reinforcement learning), belief-based algorithms, and so-called expert algorithms. An expert algorithm can be described as follows: Prior to the beginning of the game, the algorithm uses the description of the game to compute a set of experts where each expert is described as a strategy or learning algorithm that defines its behaviour in the game. The algorithms determines the highest payoff that each expert may achieve and these payoffs are then compared to an aspiration level that the algorithm considers attainable. The algorithm then identifies the experts that have the potential to achieve this aspiration and whose behaviour is compatible with a signal received, i.e. the suggestion of another player. One of those experts is then chosen that plays the game for several rounds. According to the payoffs received in these rounds, the aspiration level is updated and the experts are identified that have the potential to achieve the updated aspiration level etc.

Simulations show that under self-playing rather complex algorithms based on deep neural networks and that employ reinforcement learning like Deep-Q, Model-based Reinforcement Learning, or Win-or-Learn-Fast don't cooperate in a repeated prisoner's dilemma most of the time.[23] The most successful algorithms were the expert algorithms that are non-trivial but ultimately simple algorithmic mechanisms. It turned out that these algorithms were able to achieve cooperation much more effectively if they could send signals to each other, i.e. were allowed to communicate. Communication is modelled by a common communication protocol that contains a set of predefined messages, for example "Let's always play ⟨action pair⟩" or "I accept your last proposal", that the algorithms understand.

4.2.2 Reinforcement learning in oligopoly games

Some models of reinforcement learning in games consider not only simple two-player-two-strategy social dilemma situations but apply reinforcement learning to oligopoly models, in

[23]For a discussion of these results in the context of algorithmic collusion and antitrust, see Deng (2017).

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particular to Cournot-oligopolies. Here, the contributions by Izquierdo/Izquierdo (2015), Kaymak/Waltman (2008), Kimbrough/Lu (2003) and Kimbrough et al. (2005), and Tesauro/Kephart (2002) deserve mentioning. In these models it is demonstrated that agents may indeed arrive at a coordinated outcome. This result, however, depends on the specific details of the underlying learning process. While a simple reinforcement learning often converges to the one-shot Nash-equilibrium, under certain restrictions on the strategies available, a convergence to more coordinated outcomes was observed (Kimbrough/Lu (2003), Kimbrough et al. (2005)). Izquierdo/Izquierdo (2015) show that a simple type of reinforcement learning, the "Win-Continue, Lose-Reverse" (WCLR) rule leads to a cooperative outcome. This rule dictates an evaluation of the consequences of the last adjustment, i.e. a decrease or an increase in production, if the adjustment has led to a higher profit. Otherwise, the direction of output change is reversed. This result, however, is not robust with respect to small independent perturbations in the firms' cost or profit functions and in these cases the WCLR rule converges to the non-coordinated Cournot-Nash equilibrium. The main reason is that these small perturbations lead to a miscoordination between the firms and thus prevent a collusive outcome.

In a model of Q-learning in a Cournot oligopoly with two production levels, Kaymak/Waltman (2008) show by computer simulations, that this type of learning leads to some cooperation even if the firms have no memory, i.e. if trigger-strategies cannot be employed and even if there are more than two firms. The degree of coordination, however, is a decreasing function of the number of firms in the market. An important aspect of this model is the probability by which firms experiment, i.e. explore their environment. Cooperation results when this probability does not have a fixed value but depends on the experience of a player. A fully coordinated equilibrium is, however, in general not achieved. In a price-setting duopoly model with fixed production, where the two firms follow a Q-learning algorithm, Tesauro/Kephart (2002) find a convergence to prices that are higher than the competitive prices.

An economics paper that has recently received considerable attention in the context of algorithmic collusion is Salcedo (2015). He discusses price setting algorithms in an duopoly model with a homogeneous good. The algorithms considered are described by finite automata and don't employ reinforcement learning. The algorithm are able to perfectly monitor the history of all market outcomes. It is further assumed that from time to time, each algorithm is be able to 'decode' the algorithm used by the competitor and the own algorithm can be revised. Thus, there is perfect observability of the pricing behaviour of the opponent at each time a revision can be made.[24] This decoding has to be considered as an information exchange or communication between the two algorithms. During the periods between revisions, each firm is committed to the algorithm implemented, i.e. it cannot change its pricing behaviour. If the algorithms begin with competitive prices each, at a time of revision, the algorithm could be modified in such a way that it contains the provision that any price increase by the opponent would be matched. This information would be decoded by the competitor and its algorithm would be modified in a similar way. This modification in turn would than be observed by the first firm. As the second firm has committed for the periods between revisions, the first

[24]These assumptions are rather restrictive since deep neural networks are difficult to decode as it is not clear how and what they have learned so they may be considered as a black box. This, however, also applies to large software programs which are equally non-transparent as they contain many millions of lines of code. In addition, the firms will use private information and proprietary software. The assumption of perfect observability of the algorithm used by the competitor is therefore in general not satisfied in real markets (see Oxera (2017), 19).

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algorithm would set a higher price that would be matched by the second firm etc. This renegotiation channel and the (imperfect) commitment, i.e. the opportunity to revise an algorithm after some periods, preclude equilibria with low prices. Firms would increase their prices over time until the prices are realized that maximize the joint profit. It can easily be seen that in this model, collusion is the only outcome. It has to be noted, however, that this is in fact not a model of tacit collusion as the title of the paper suggests, but contains, via the decoding of an algorithm, explicit communication between the firms and thus is equivalent to an explicit collusion between the firms. If this type of information exchange would not exist, i.e. if there were no communication between the algorithms, the collusive outcome would not emerge.

This brief review of some of the contributions from the literature on algorithmic collusion shows that a coordinated behaviour of algorithms is a possible outcome, but it is not as quick and easy or even unavoidable as it is often assumed in the legal discussion of algorithmic collusion. There are several conclusions that may be drawn from this literature. First, whether cooperation is achieved depends very much on the specific details of the environment and also on the algorithms employed. Often, cooperation can be achieved with rather simple mechanisms while complex algorithms based on deep neural networks and employ methods of deep learning often fail to cooperate. In practice, a wide variety of pricing and learning algorithms is employed, e.g. bandit-type models, customer choice models, econometric regression models, machine learning models, and greedy ad-hoc approaches which exhibit substantial variation in performance, depending on market structure and market dynamics.[25] In addition, many models assume that the environment is fixed while in real world market often changes occur, for example market entry or exit, mergers, innovations etc. If for example, a new product is introduced in a market, the demand system, including all substitution relationships between the goods will change and this modified demand has to be learned anew by the algorithms. Further, the environments considered are rather simple as compared to real world markets. For example, cooperation is generated in simple games with just two actions (cooperate and defect) while in a market, multi-dimensions actions, involving prices, quality, service etc. can be chosen. A collusive outcome implies that coordination is achieved over all main competitive variables. Otherwise, in case coordination on price only, the firms could still compete with the quality of the products or the service offered. Also, algorithmic cooperation is often tested with self-play only and it remains unclear what would happen if different algorithms played against each other. The research in computer science on algorithmic collusion thus raises serious doubts whether algorithmic collusion in markets is likely to happen. Nevertheless, it cannot be excluded that algorithms might learn over time to coordinate their behaviour and achieve a collusive outcome.

As price setting algorithms have become much more complex in the last couple of years, studies of human behaviour in situations of strategic interdependence might shed some light on the question under which conditions coordinated and collusive behaviour of algorithms is to be expected. In the following section the conclusions that can be drawn from the computer science literature are compared with results from experimental economics about tacit and explicit collusion in oligopolistic markets.

[25] See van de Geer et al. (2018).

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5 Collusion in oligopoly - Results from experimental economics

As in the literature on algorithmic collusion the focus here is on tacitly collusive behaviour, i.e. collusion without any explicit agreement or communication between the firms, we consider in this section first oligopoly experiments on tacit collusion. The problem of tacit collusion in oligopoly has been studied in experimental economics for decades. In what follows we give a brief survey of the main results from this research on tacit collusion to look at the main factors that are important for a collusive outcome to emerge. We will concentrate mainly on several meta-studies on oligopolistic collusion (Engel (2015), Haan et al. (2006), Horstmann et al. (2016), Potters/Suetens (2014)). Secondly, the impact of communication between participants on collusive outcomes is considered and some studies that deal with number effects and the role of communication are discussed. The experiments have indicated that communication is of major importance for a collusive outcome to be achieved in particular in the case of more than two firms. This indicates that the ability to communicate could be also of importance in the context of algorithmic collusion.

5.1 Number effects

The most important aspect with respect to collusion is the number of competitors in the market. This has been demonstrated in a study of number effects in experimental oligopolies (Huck et al. (2001)) who examined oligopolies with two, three, four, and five firms. They found some collusion with two firms while three-firm oligopolies tend to produce competitive outputs and oligopolies with four or more firms never reach a collusive outcome. Potters/Suetens (2014) in their survey of the recent literature of experimental oligopolistic markets come to a similar conclusion: "...implicit coordination on a joint-profit maximizing price is frequently observed in markets with two sellers, rarely in markets with three sellers, and almost never in markets with four or more sellers."[26] This is basically confirmed in the meta-analysis of Horstmann et al. (2016): "Within and across the surveyed oligopoly experiments, markets with two firms are significantly more prone to tacit collusion than markets with three as well as four firms, everything else being equal."[27] Also Engel (2015) shows that experimental studies in general find that collusive outcomes depend strongly on the number of participants. These results hold for quantity as well as for price competition.

5.2 Collusion and communication

But not only the number of participants is relevant for a coordinated behaviour. Several studies have pointed out that communication between agents is an important determinant of whether a collusive outcome can also occur in markets with more than two players. As in repeated games there are usually infinitely many Nash equilibria, non-binding communication between agents makes it much easier for players to coordinate on a specific collusive outcome. This has already been pointed out by Friedman (1967) who found that market prices are significantly higher if players can communicate by sending non-binding messages, than without

[26] Potters/Suetens (2014), 448.
[27] Horstmann et al. (2016), 9.

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communication.[28] This is also the result of the meta study by Engel (2015): "If participants are additionally allowed to communicate price or quantity choices, collusion increases substantially and significantly..."[29] The importance of communication for a collusive outcome has not only been demonstrated in experimental oligopolies but also in a case study of the Airline Tariff Publishing Case (Borenstein (1999)). Here, the ability to quickly exchange price information was an important factor in achieving a collusive outcome.

In a recent study, Fonseca/Normann (2012) examine how the impact of communication changes with the number of participants in a market. They find that the impact of being able to communicate on profits is non-monotonic in the number of firms in a market. While in duopolistic markets, communication is in general not necessary for a collusive outcome, in large markets a collusive outcome does not emerge even if the firms are able to talk. In markets with a medium number of firms, however, the ability to communicate may drastically change the outcome: Without communication, the market outcome is competitive while with communication firms are able to maintain at least some degree of coordination, i.e. prices are higher than under competition. A similar result has been obtained by Harrington et al. (2016). They found that non-binding price announcements led to a collusion only in the duopoly case with symmetric firms. When subjects were allowed to communicate unrestrictedly, they coordinated on high prices even if they were asymmetric.

This indicates that the ability to communicate is an important factor to establish a collusive outcome in experimental oligopolies.[30] It may therefore be conjectured that this holds also in the case of algorithmic collusion and would imply that collusion depends very much on the ability of the agents or algorithms to communicate. This is of particular importance in the context of the claim made in the legal literature that algorithmic coordination leads to collusive behaviour even in markets with a medium or large number of firms. The fact that communication has a strong and positive effect also in the case of algorithmic collusion has already been pointed out in several contributions in the computer science literature, e.g. the papers by Tan (1993) or Crandall et al. (2017) and also implicitly in the paper by Salcedo (2015). Thus, the question arises whether algorithms are able to communicate with each other, or, whether different algorithms might even be able to learn to communicate without being explicitly programmed, i.e. without a common communication protocol. This is a topic that has recently found some attention in the literature on machine learning and coordinated behaviour of algorithms but also in the literature on robotics. Therefore, the question whether algorithms are able to learn to communicate is considered in section six.

6 Can algorithms learn to communicate?

The question of communication between algorithms has found some attention in the computer science and machine learning literature because many problems in artificial intelligence require the cooperation between multiple agents. In general, the problem of communication has been dealt with by communication protocols that were fixed and that the agents could

[28] There is also a considerable theoretical literature that examines the role of communication for collusive outcomes in repeated games which is not discussed here. See e.g. Aoyagi (2005), Awaya/Krishna (2015), Compte (1998), Garrod/Olczak (2017), Kandori (2003), Kandori/Matsushima (1998), Obara (2009), Spector (2015), Tomala (2009).

[29] Engel (2015), 19.

[30] See Haan et al. (2006), 24.

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use to exchange messages as for example in the paper by Crandall et al. (2017). Also, most contributions study learning of communication in a fully cooperative setting, i.e. a scenario without any tension between individual and collective rationality and thus different from the framework of collusion in oligopolies.

An early paper on learning to communicate in a multi-agent setting is Giles/Jim (2002). The authors consider a predator-prey environment were the predators communicate simultaneously to a message board. Learning to communicate is modelled by a genetic algorithm. Simulations showed that the evolved communicating predators were able to perform significantly better than predators that were not able to learn to communicate. This indicates that the system was able to learn superior communicating strategies that were not designed by the programmer. In a similar setting, Kasai et al. (2008) consider multi-agent reinforcement learning with communication between the agents where the agents obtain communication codes through learning. Communication codes are bit sequences of variable length, ranging from 0 to 4 bits. When agents learn 1 or more bits, the results with communication outperformed the outcomes in the case without communication.

In the context of learning to communicate, Foerster et al. (2016) consider the framework of fully cooperative partially observable, sequential multi-agent decision problems. Agents maximise the same discounted sum of rewards and each agent receives a private observation about the environment. Each agent can take actions that affect the environment and can also communicate with its fellow agents. The agents have to learn a communication protocol that enables them to coordinate their behaviour and solve the task. Here, the learning occurs in a centralized way, i.e. during learning which is performed by a centralised algorithm, communication between agents is not restricted. During execution of the learned policies, however the agents can communicate only via a channel with limited bandwidth. The authors consider two different types of communication learning, one based on deep Q-learning, the other allows for error correction (backpropagation) through communication channels. The simulation results show that the learning method that relies on error correction using backpropagation outperforms the other method. The authors, however, consider their results only as a "first attempt at learning communication and language with deep learning approaches."[31]

In a recent paper, Sukhbaatar et al. (2016) consider a fully cooperative setting were the agents have only limited information about the environment and have to cooperate to perform a complex task. They consider a neural model to analyze communication learning. This is applied to a variety of tasks, for example a lever pulling task consisting of m levers and m randomly drawn agents from a population. The agents must simultaneously choose a lever to pull. The goal is for each of them to pull a different lever as the payoff of each agent is proportional to the number of distinct levers pulled. In this framework, simulations showed that agents learn to communicate amongst themselves, yielding improved performance. In some cases, the language devised by the agents can be interpreted, revealing simple but effective strategies for solving the respective task.

A related problem has recently been studied in the robotics literature (Barrett et al. (2017)). The starting point of this analysis are robots produced by different companies which don't share a common communication or coordination protocol. Even without a standardized communication protocols, robots may potentially have to cooperate with a variety of teammates

[31] Foerster et al. (2016), 9.

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in order to achieve a shared goal, i.e. an ad hoc teamwork is required. The main problem is to recognize that a key requirement of this teamwork is the adaptability to previously unknown agents. The authors have developed a general purpose algorithm that was able to identify and to learn to adapt to and to cooperate with new teammates.

An interesting recent experiment by Abadi/Anderson (2016) from the Google Brain project has shown that algorithms can learn to hide their communication from third parties. They considered two neural networks that learn to encrypt messages they send to each other such that a third neural network is limited in what it can learn from eavesdropping of the conversation between the first two networks. It is demonstrated that neural networks are indeed able to learn how to encrypt and decrypt messages, and also how to apply these operations selectively in order to meet confidentiality goals.

To sum up, the development of algorithms that are able to learn to communicate with each other seems to be in its very early stages. It is unclear which types of communication between algorithms might arise.[32] Presently, we should not expect that different pricing algorithms will be able to communicate with each other or, as suggested by Salcedo (2015), to be able to 'decode' other algorithms and to achieve a collusive outcome. Taking into account the current state of research in artificial intelligence, machine learning, and deep learning as well as cooperation and communication between algorithms, the statement by Ezrachi/Stucke: "Two Artificial Neural Network and one Nash equilibrium meet in an online (pub) hub. After a few milliseconds, a unique silent friendship is formed . . . " (Ezrachi/Stucke (2017)) does not seem to describe the actual situation.

Nevertheless, it cannot be excluded, that – taking into account the rapid progress in AI research – algorithmic collusion might occur in the future and that algorithms may learn to communicate and thus, that the likelihood of algorithmic collusion increases. Therefore, the question has to be considered how to deal with these problems from a competition law and competition policy point of view.

7 Algorithmic collusion and competition policy

Until today, there is no case where autonomous agents have coordinated their price setting behaviour to maximize their joint profits, i.e. to tacitly collude. There are some cases however, where algorithms have been used as facilitating devices like a prepaid mobile phone, a pricing formula or a meeting competition clause to make collusion more effective.[33]Here, the formation of a cartel was a concious and deliberate decision by the managers of the participating firms. As in these cases the term 'algorithm' can be replaced with the expression 'a guy named Bob', as FTC Commissioner Ohlhausen has put it, no novel competition problems arise and the existing instruments of competition policy and competition law seem sufficient to deal with these issues.

With respect to self-learning, autonomous algorithms, the problem is a different one because in

[32]In a recent experiment carried out by the Facebook artificial intelligence research division, two artificially intelligent chatbots were challenged to negotiate with each other over a trade. The chatbots, however, made changes to the English they used during the negotiations and developed their own language, a type of shorthand. While the chatbots seemed to understand the language and carried out transactions using only this language, it was incomprehensible to the human experimenters. The experiment was thus aborted. See Griffin (2017).

[33]For example, the Topkins or the Trod case.

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these cases, no humans are directly involved in forming a tacitly collusive outcome - there was no agreement or meeting of the minds in the usual sense and the firms might not even have intended this outcome. Their intention might have been to use an efficient and competitive pricing mechanism and collusion was an outcome that may have neither been planned nor foreseen by the firms but has been learned by the algorithms on their own. In those cases the question arises whether a tacitly collusive behaviour of price setting, self-learning algorithms is a violation of competition law at all. Under the current competition law, this is in general not the case.[34] Thus, one option to deal with this problem is to extend the meaning of "agreement" to include also cases of algorithmic collusion. Similar problems have been discussed in the context of purely tacit collusion in oligopoly. Posner for example has suggested that tacit collusion should be considered in the same way as an explicit collusion:

> "If seller A restricts his output in the expectation that B will do likewise, and B restricts his output in a like expectation, there is a literal meeting of the minds, a mutual understanding, even if there is no overt communication. In forbearing to seek short-term gains at each other's expense in order to reap monopoly benefits that only such mutual forbearance will allow, A and B are like the parties to a "unilateral contract" which is treated by the law as concerted rather than individual behaviour." (Posner (1976), 71f.)

But even if algorithmic collusion would be considered as a violation of competition law, the problem arises how to verify whether algorithms have in fact colluded, i.e. whether the algorithms have converged at collusive prices. For a close examination of a case of algorithmic collusion, competition authorities must have an initial suspicion that such a collusive behaviour has indeed occurred. For this purpose, competition authorities themselves could employ algorithms that observe market prices to detect any signs of collusive behaviour.[35] If there is a reasonable suspicion that there is an algorithmic collusion, the problem of the burden of proof has to be addressed. Should companies that employ price setting algorithms be required to explain the functioning of their algorithms in a comprehensible way to the competition authorities and the courts? This requirement would implicitly restrict the class of algorithms that can be employed as for example, the functioning of a deep neural network would be hard to explain to third parties. Or is the burden on the competition authorities who have to demonstrate that the algorithms used by the firms have indeed caused a collusive market outcome? In either case, competition authorities need sufficient in-house or third-party expertise in computer science and in particular in artificial intelligence to assess the claims.

If it has been verified beyond reasonable doubt, that the algorithms have indeed converged to a collusive equilibrium and if this is a potential violation of competition law, the question of liability arises. Is the programmer of the algorithm liable or is it the owners of the firm? That may depend very much on the specific regulations in the respective jurisdiction. Under European law, external programmers or software service companies would generally not be considered as liable for any violation of competition law since programmers work mainly according to the instructions and specifications of the company that implements the algorithm. Its the company that is liable for any acts of individual entitled to act on behalf of the company.[36]

[34] Salaschek/Serafimowa (2018), 12.

[35] Smart algorithms, however, could make it very difficult for observers to detect any collusive behaviour by slightly changing the inflated prices at random so that this price variability would appear as normal competitive behaviour.

[36] Salaschek/Serafimowa (2018), 16.

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In this context it is irrelevant whether self-learning algorithms are classified as individuals or if they are considered as the extended arm of individuals who employ these algorithms.[37] But this may be different in other jurisdictions.

To avoid the problem of algorithmic collusion altogether, would be to prohibit the use of self-learning price-setting algorithms in the first place. This, however does not seem an acceptable solution because this would also preclude the realization of many efficiencies that these algorithms generate. Also, the suggestion by Ezrachi/Stucke: "...to condemn and challenge the creation of market conditions which led to sustaining tacit collusion – the creation of a transparent market in which monitoring and punishment mechanisms are present"[38] is not convincing. First, it is in general difficult to assess whether the structural market conditions for tacit collusion are satisfied and secondly, it is unclear what competition authorities could do to challenge the creation of a transparent market. In this context, it has to be taken into account that market transparency is advantageous for consumers because they can easily compare prices and find the best offer.

In a recent paper on the problem of competition law and autonomous price setting agents, (Harrington (2017)) suggests to deal with the problem of liability by restricting the class of allowable algorithms or by prohibiting algorithms with certain features that support prices above the competitive level. Harrington suggests a three-step procedure, to analyze which types or properties of algorithms should be prohibited. This approach is based on a simulated market, where, under a variety of market conditions, the properties of algorithms that lead to a collusive outcome are determined. This simulation would have to be carried out by the competition authorities or a specialized agency. Such an auditing of algorithms however, is a gragantuan task as there are so many different algorithms in use which are constantly either modified by the programmers or by self learning and which might develop new, not yet prohibited properties that could lead to a collusive outcome. The examination of algorithms would thus have to be carried out regularly.

Finally, one could consider the option of incorporating legal provisions and constraints at a fundamental level of each algorithm, similar to the three laws of robotics proposed by I. Asimov. For example, only algorithms are accepted that are unable to communicate. As algorithms observe the pricing behaviour of their competitors they could learn to communicate by sending messages encoded in the prices charged, similar to the code bidding that has been observed in procurement auctions. Thus, the problem arises whether sufficiently sophisticated algorithms may be able to circumvent these provisions.

All in all, the options for competition law and competition policy to deal with the potential problem of algorithmic collusion seem to be rather limited. Only if the concept of an agreement is extended to cover also the case of this type of collusive behaviour, it could be considered as a violation of competition law at all. But even then, severe problems remain with respect to the detection as well as the verification of algorithmic collusion. In addition, the suggestions how to prevent it, put a heavy burden on competition authorities as this would require considerable expertise in computer science as well as significant additional resources to audit the many different and constantly changing price setting algorithms that are employed by firms.

[37] Käseberg/von Kalben (2018), 5.
[38] Ezrachi/Stucke (2015), 21.

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8 Conclusion

The discussion of the computer science literature on algorithmic collusion as well as the studies of collusion in experimental oligopolies has shown that this type of collusive behaviour is not as likely or even unavoidable as the some legal scholars seem to suspect. One cannot completely rule out the possibility that algorithms might be able to tacitly collude their price setting behaviour and to charge inflated prices that maximize their joint profit, but collusive behaviour of algorithms is significantly more complex than it is suggested in many contributions to that issue.

If algorithms become as complex as humans in certain highly specialized areas as for example in games as Chess or Go, problems similar to those that have been demonstrated in many studies of collusion in experimental oligopolies may arise. These experiments have shown that without any communication between the participants, collusion generally takes place in duopolies. This suggests that algorithms may have to learn to communicate with each other to achieve a collusive outcome in particular in market with several participants. This first steps in this direction have been made in the artificial intelligence literature, but currently, it seems that there is still a long way to go until algorithms are developed that are able to learn to communicate without any predefined communication protocol.

Even if algorithms were able to coordinate their price-setting behaviour, it is by no means clear how this problem should be dealt with by competition policy and competition law. Even if competition law would be modified to include algorithmic collusion as an infringement, its detection and verification appears to be fraught with considerable difficulties. This also holds true for the problem of preventing algorithmic collusion in the first place. If the use of self-learning pricing algorithms is not prohibited altogether, algorithms have to be regularly monitored which would put a considerable burden on competition authorities.

Give that the problem of algorithmic collusion at the moment rather belongs to the realm of legal sci-fi, it seems advisable to devote the limited resources of competition authorities to more pressing competitive concerns than these highly speculative conjectures about algorithmic collusion, in particular to investigation of potential abuses of dominant market positions, in particular of large online-platforms.

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